

Holly Mandel

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Machine learning engineer and mathematics PhD with experience in quant finance and Python software engineering

EDUCATION

University of California, Berkeley 2017–2022

PhD, Mathematics

Degenerations of Negative Kähler-Einstein Surfaces ([J London Math Soc](#), high-tier generalist journal)

Yale University 2010–2014

BS, Biology

Phi Beta Kappa, *summa cum laude* (top 5%), distinction in the major

EMPLOYMENT

AdeptID, Machine Learning Engineer October 2024–Present

- Created dense retrieval system for job candidate search tool that improved recall by 60%
- Designed and tested LLM-driven translation system for user documents to achieve 99.99% reliability
- Designed validation framework (Pydantic, Pandera) for core models that dramatically improved research efficiency

Vatic Labs, Quantitative Researcher June 2022–May 2024

- Developed statistical arbitrage trading strategy with Sharpe >2.0
- Designed and implemented an automated strategy selection pipeline that improved Sharpe at scale relative to existing rules-based method

Clark lab, Yale MCDB, Research Associate June 2014–August 2015

- Published first measurement of third-order kernels representing fruit fly motion vision ([Neuron](#))
- Demonstrated functional relationship between nonlinearities in fly motion detection and natural scene statistics ([eLife](#))

OTHER EXPERIENCE

XArray, Open source contributor August 2024–October 2024

- Updated interpolation routines to reflect changes in underlying numerical libraries (4 issues resolved)

SKILLS

Languages: Python, C++ (intermediate), MATLAB, $\text{\LaTeX}$

ML / Data: PyTorch, Hugging Face, Pandas, Xarray, LLMs, Coiled, Optuna

Systems: Git, Docker, AWS (S3, ECS, OpenSearch), MongoDB

Testing / Tooling: Pytest, Pydantic, Pandera, test-driven development

FELLOWSHIPS AND PRIZES

Eliciting Latent Knowledge Contest prizewinner, Alignment Research Center, joint with Jacob Hilton 2022

Graduate Research Fellowship, NSF 2017

Presidential Fellowship, Academic Excellence Award for top score on mathematics qualifying exam,
Rutgers University 2016

William R. Belknap Prize for Academic Excellence in Biology, Yale Department of MCDB 2014